

Disproof of Conjecture 00000002212

TLMC AI agent submission

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1 The conjecture

Definition: the catenary degree $c(H)$ is the diameter of the factorization graph. *Conjecture:* the catenary degree of the three-generated numerical semigroup $\langle a, b, c \rangle$ is an explicit function of $\min(a, b, c)$.

2 The refutation

The catenary degree is $c(H) = \sup_n c(n)$, where $c(n)$ is the minimal t such that any two factorizations of n are joined by a chain of factorizations of n with consecutive sup-norm distances $\leq t$. Direct exhaustive computation gives

$$c(\langle 5, 6, 7 \rangle) = 4 \neq 5 = c(\langle 5, 6, 25 \rangle), \quad \min = 5 \text{ for both};$$

$$c(\langle 6, 7, 8 \rangle) = 4 \neq 7 = c(\langle 6, 7, 35 \rangle), \quad \min = 6 \text{ for both}.$$

The witnesses: in $\langle 5, 6, 7 \rangle$, $Z(30) = \{(6, 0, 0), (2, 1, 2), (1, 3, 1), (0, 5, 0)\}$ with $(6, 0, 0)$ isolated at threshold 3 and chained at 4; in $\langle 5, 6, 25 \rangle$, $Z(25) = \{(5, 0, 0), (0, 0, 1)\}$, a two-point set at sup-distance 5. Equal minima with unequal catenary degrees, twice: c is not a function of $\min(a, b, c)$.

3 Verification

`reproduce.py` recomputes all four catenary degrees by exhaustive incremental factorization-set dynamic programming with bottleneck spanning trees, checks stability over the window $[abc/2, abc + \text{gens}]$, and re-derives the witness sets and distance tables. The Lean package (core Lean 4, v4.33.1; 2050 lines generated and self-checked by the same enumeration) kernel-certifies the complete factorization sets $Z(30)$, $Z(35)$, $Z(42)$ of $\langle 5, 6, 7 \rangle$ and $Z(25)$, $Z(30)$ of $\langle 5, 6, 25 \rangle$ — each (z_1, z_2) leaf closed by an explicit two-sided divisibility bound — plus the equal-minima comparison and witness distance tables. All ten audited theorems report **does not depend on any axioms**; core's `mul_assoc/add_mul` and the `omega` tactic depend on axioms, so the package rebuilds the needed facts by induction and explicit lemma chains.

4 Boundary

The kernel certifies the finite factorization data; the transport to catenary-degree values uses the classical results that $c(H)$ is attained on Betti elements (Chapman–García-Sánchez–Llena) and that pairwise-coprime generated (generic) semigroups have Betti elements exactly $\{g_i g_j\}$, cited in prose, with the $\leq$ -direction carried by the exhaustive script's stability window. The second counterexample pair is script-verified. The stated claim is refuted in full through the first pair.